STUDENT NUMBER:

APSC 171 - Term Test 1
Tuesday October 3, 2023

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Student #: 24601

- This is a 2 hour test.
- No textbooks, notes, or aids other than your calculator (Casio 991 series) are allowed.
- For full marks, you must show all your work and explain how you arrived at your answers, unless explicitly told to do otherwise.
- Wherever appropriate, **include units in your answers**.
- When drawing graphs, **add labels and scales on all axes**.
- Correct final answers with incorrect support may be given no credit.
- While waiting to start, fill in your name and student number above.
- If you need more room, there are blank pages at the end of the test. **If you use these pages, you must provide clear directions on the original page**, e.g. "Continued on page 10".
- Only answers written in this booklet will be graded; all scrap paper from the test will be discarded.

Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	Q 7	Q 8	Total
8	8	9	8	8	5	9	5	60

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1. Find the derivative of the following functions. You may assume that the input variables will always be within the domain of their respective functions.
You do **not** need to simplify after you have taken the derivative.

(a) $f(x) = 3x^2\cos(4x)$. [/2]

$f'(x) = 6x\cos(4x) - 3x^2\sin(4x)\cdot 4$

$= 6x\cos(4x) - 12x^2\sin(4x)$

(b) $g(t) = \dfrac{e^{at}}{1-e^{bt}}$. Both a and b are constants. [/3]

$g'(t) = \dfrac{ae^{at}(1-e^{bt}) - e^{at}(-be^{bt})}{(1-e^{bt})^2}$

(c) $h(x) = \ln\left(3\sqrt{x^2+1}\right)$. [/3]

$h'(x) = \dfrac{1}{3\sqrt{x^2+1}} \cdot 3\cdot\frac{1}{2}\dfrac{1}{\sqrt{x^2+1}}\cdot 2x$

$= \dfrac{x}{x^2+1}$

Alternative:
simplify first.

$h(x) = \ln(3) + \frac{1}{2}\ln(x^2+1)$

$h'(x) = \frac{1}{2}\dfrac{1}{x^2+1}\cdot 2x$

$= \dfrac{x}{x^2+1}$

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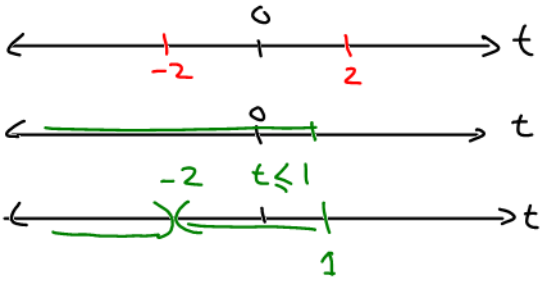
2. The following three parts are independent of each other.

(a) Find the domain of the function $\vec{r}(t) = \left\langle 2\ln(t^2+1),\ \frac{2t}{t^2-4},\ \sqrt{1-t} \right\rangle$.
Express the domain as an interval, or a union of intervals. E.g. "$t \in [-2, 2)$", or "$t \in (-\infty, 2) \cup [3, \infty)$". [/2]

x domain: $t^2+1 > 0$ → always true.

y domain: $t^2-4 \neq 0$ → $t \neq 2, -2$

z domain: $1-t \geq 0$ → $t \leq 1$



Combined: $t \in (-\infty, -2) \cup (-2, 1]$

(b) Compute the angle between the vectors $\vec{a} = \langle -2, 5, 1\rangle$ and $\vec{b} = \langle 2, 1, 2\rangle$.
Your answer should be in radians, and in the range $[0, \pi]$; report your answer to 4 significant figures. [/2]

$\vec{a}\cdot\vec{b} = -4+5+2 = 3$

$\|\vec{a}\| = \sqrt{4+25+1} = \sqrt{30}$

$\|\vec{b}\| = \sqrt{4+1+4} = \sqrt{9} = 3$

$\vec{a}\cdot\vec{b} = \|\vec{a}\|\,\|\vec{b}\|\cos\theta$

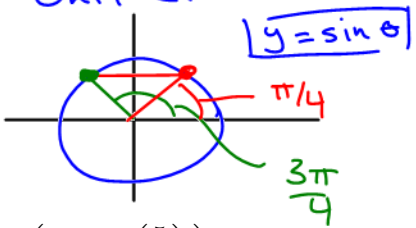
subbing in, $3 = \sqrt{30}\cdot 3\cdot\cos\theta$ → $\cos\theta = \frac{1}{\sqrt{30}}$ → $\theta = 1.3872$ rads

(c) Find the exact value of $x = \arcsin\left(\sin\left(\frac{11\pi}{4}\right)\right)$. [/2]

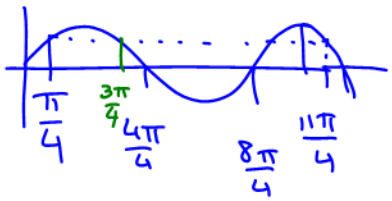
$x = \frac{\pi}{4}$

or $-2\pi : \frac{3\pi}{4}$

Unit Circle



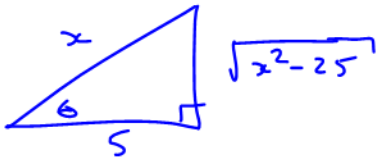
Sine Graph



(d) Express the relationship $y = \tan\left(\arccos\left(\frac{5}{x}\right)\right)$ without trigonometric functions. [/2]

$\theta = \arccos\left(\frac{5}{x}\right)$

$y = \tan\theta = \frac{\sqrt{x^2-25}}{5}$



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3. Consider a particle whose trajectory is defined by $\vec{s}(t) = \langle 3\sin(2t) + 2, \ -3\cos(2t) + 4 \rangle$.

(a) The particle's trajectory lies on one of the circles defined by the Cartesian equations below:

- Circle A: $(x+2)^2 + (y+4)^2 = 9$
- Circle B: $(x+2)^2 + (y-4)^2 = 9$
- Circle C: $(x-2)^2 + (y-4)^2 = 9$

Circle the correct formula and then support your answer with an appropriate calculation. [/2]

$LHS = (x-2)^2 + (y-4)^2$

$= (3\sin(2t) + 2 - 2)^2 + (-3\cos t + 4 - 4)^2$

$= 9\sin^2(2t) + 9\cos^2(2t) = 9 = RHS$ ✓

(b) Find the velocity vector for the particle's trajectory. [/2]

$\vec{v}(t) = \frac{d}{dt}\vec{s}(t) = \langle 6\cos(2t), 6\sin(2t) \rangle$

(c) Find the velocity vector for the particle specifically at $t = \frac{\pi}{3}$, then compute the **unit** vector pointing in the direction of the velocity at that time. [/2]

At $t = \frac{\pi}{3}$, $\vec{v}(\pi/3) = \langle 6\cos(2\cdot\frac{\pi}{3}), 6\sin(2\cdot\frac{\pi}{3}) \rangle$

$= \langle 6(-\frac{1}{2}), 6(\frac{\sqrt{3}}{2}) \rangle$

$= \langle -3, 3\sqrt{3} \rangle$

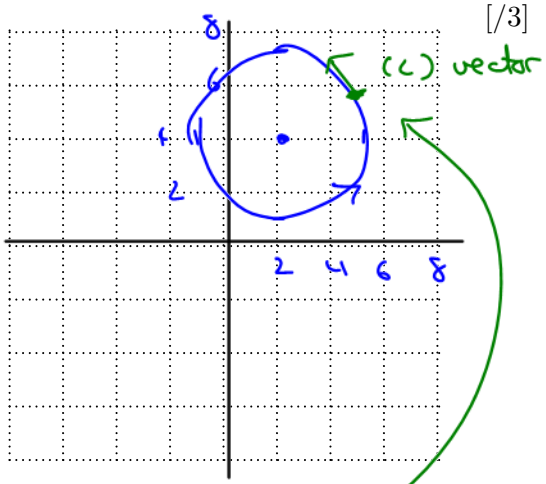
unit vector $= \frac{1}{\sqrt{3^2 + (3\sqrt{3})^2}} \langle -3, 3\sqrt{3} \rangle$

$\approx \langle -0.5, 0.866 \rangle$

unit vector = $\langle -0.5, 0.866 \rangle$

(d) On the grid to the right, draw x and y axes then sketch the graph of the trajectory. [/3]
Once you have drawn the graph:

- Draw arrows on the trajectory to indicate the direction of motion;
- Add your answer to part (c) at the appropriate location on the trajectory.



position at $t = \pi/3$

$= \langle 3\sin(2\cdot\frac{\pi}{3}) + 2, -3\cos(2\cdot\frac{\pi}{3}) + 4 \rangle$

$\approx \langle 4.59, 5.5 \rangle$

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4. The following three parts are independent of each other.

(a) Find a value s for which the vectors $\vec{a} = \langle s,\ s-1\rangle$ and $\vec{b} = \langle s^2,\ s\rangle$ will be parallel. [/3]

Parallel → scalar multiples

⇒ x: $s = cs^2$ → $s=0$ (no solution) or $1 = cs$ → $c = \frac{1}{s}$ (1)

y: $s-1 = cs$ (2)

(1) → (2) $s - 1 = \left(\frac{1}{s}\right)(s) = 1$

$\boxed{s=2}$

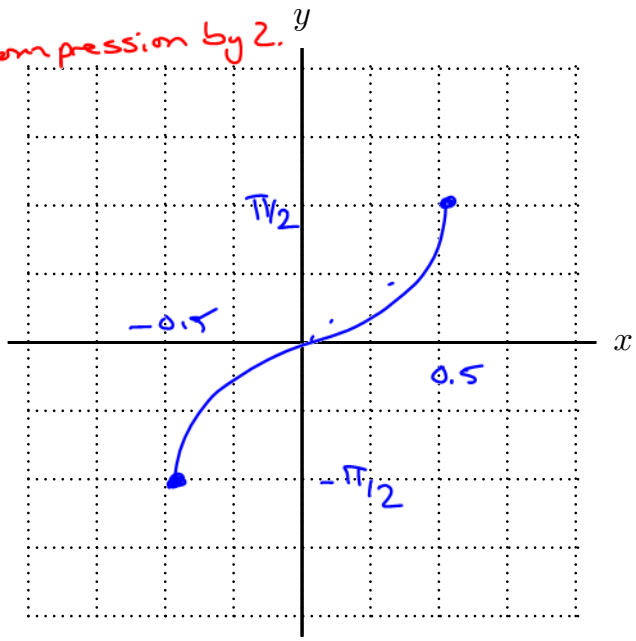
Check: for $s=2$, $\vec{a} = \langle 2, 1\rangle$ and $\vec{b} = \langle 4, 2\rangle = 2\vec{a}$ ✓

(b) On the axes to the right, sketch the graph of the function $y = \arcsin(2x)$. [/3]

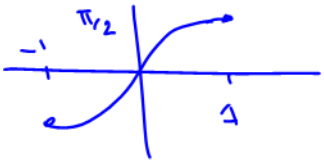
- Use filled circles for endpoints that are reached on the graph (end of a closed interval), and
- open circles for endpoints that are not reached (end of an open interval).

Clearly indicate the scale on your axes

horiz'l compression by 2.



$y = \sin x$



(c) Compute the derivative of $f(x) = \arctan(4x+5)$. [/2]

$$f'(x) = \frac{1}{1+(4x+5)^2} \cdot 4$$

$$= \frac{4}{16x^2+40x+26}$$

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5. The position of a particle is defined by the vector-valued function $\vec{r}(t) = \langle t^2 - 1,\ 2 - t \rangle$.

(a) Eliminate parameters to find a Cartesian equation for the function $\vec{r}(t)$. [/3]

① $x = t^2 - 1$

② $y = 2 - t$

② → $t = 2 - y$ → ① $\boxed{x = (2-y)^2 - 1}$

or

$t^2 = x + 1$

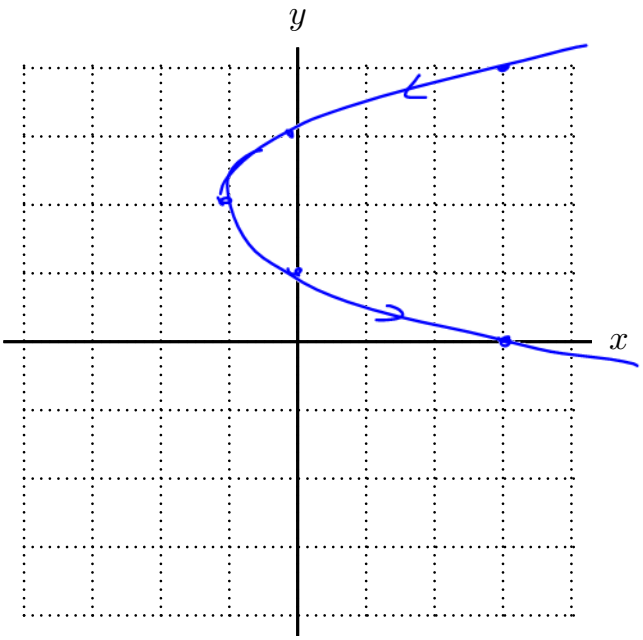
$t = \pm\sqrt{x+1}$

↪ ② $\boxed{y = 2 \pm \sqrt{x+1}}$

(b) On the axes to the right, sketch the graph of the Cartesian equation you found in part (a). Once you have drawn the graph: [/3]

- Clearly indicate what part of this graph the function $\vec{r}(t)$ actually passes over, and
- draw arrows to indicate the direction of motion.

$x = (2-y)^2 - 1$



x	y
3	0
0	1
−1	2
0	3

- as t incr, y decr
- domain is all t values.

(c) Find a formula for the acceleration of the particle at time t. [/2]

Need velocity first: $r'(t) = \langle 2t, -1 \rangle$

Then acceleration: $\boxed{r''(t) = \langle 2, 0 \rangle}$

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6. This question involves the same vector-valued function as the previous question:

$$\vec{r}(t) = \langle t^2 - 1,\ 2 - t \rangle.$$

For what value(s) of t does a line tangent to the trajectory pass through the point (-4 ,2)? [/5]

velocity $\vec{v}(t) = \langle 2t, -1 \rangle$

$\Rightarrow$ Tangent line @ $t=a$ will be

$\vec{L}(t) = \underbrace{\langle a^2-1, 2-a \rangle}_{\vec{r}(a)} + (t-a)\underbrace{\langle 2a, -1 \rangle}_{\vec{v}(a)}$

$\vec{L}(t) = \vec{r}(a) + (t-a)\,\vec{v}(a)$

Note: using $\vec{r}(a) + (t)\,\vec{v}(a)$ will also work and is acceptable.

Set equal to $\langle -4, 2 \rangle$ in x and y:

x: $-4 = a^2 - 1 + (t-a)(2a)$ ①

y: $2 = 2 - a + (t-a)(-1)$ ②

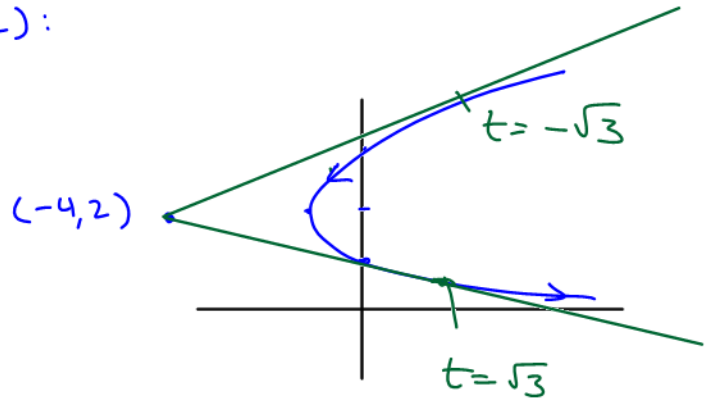
Tidying ② $\rightarrow$ $0 = -a - t + a \rightarrow \boxed{t = 0}$

$t=0 \rightarrow$ ① $\quad -4 = a^2 - 1 + (-a)(2a)$

Tidying $\quad -3 = -a^2$

$\boxed{a = \pm\sqrt{3}}$

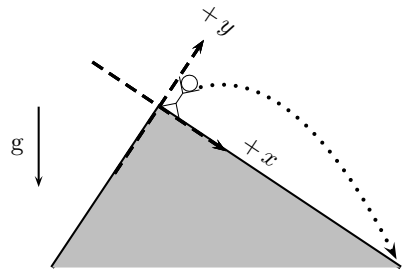
At the points $t = -\sqrt{3}$ and $t = +\sqrt{3}$, the tangent line will pass through $(-4, 2)$:



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7. You throw a ball down the side of a hill. Relative to the xy coordinate axes shown in the diagram, the velocity of the ball at time t (in seconds) is given by

$$\vec{v}(t) = \langle \underbrace{6t+3}_{v_x(t)}, \underbrace{-12t+5}_{v_y(t)} \rangle \text{ m/s.}$$



(a) Give a function $\vec{r}(t) = \langle x(t), y(t) \rangle$ which describes the position of the ball at time t. Assume that the ball is thrown from $\langle x, y \rangle = \langle 0, 0 \rangle$ at $t = 0$. [/3]

$\vec{r}(t) = \langle \frac{6t^2}{2} + 3t + x_0, -\frac{12t^2}{2} + 5t + y_0 \rangle$

Since $x(0)=0$ and $y(0)=0$, $x_0 = 0$ and $y_0 = 0$

$\Rightarrow \quad r(t) = \langle 3t^2+3t, -6t^2+5t \rangle$

(b) At what time t will the ball hit the hillside? [/2]

Landing $\Rightarrow y=0$ $\quad -6t^2+5t=0$

$t(-6t+5)=0$ $\quad t=0$ (launch)

or $-6t+5=0$

$t = 5/6$ (landing)

(c) Since there is no wind, the only acceleration on the ball is due to gravity. Give a vector in xy coordinates which points in the direction of gravity. [/2]

$\vec{a} = \vec{v}' = \langle 6, -12 \rangle \quad m/s^2$

(d) The gravity on Earth, Mars and Neptune are 9.8m/s^2, 3.8m/s^2 and 13.3m/s^2 respectively. Which planet are we on? Be sure to justify your answer. [/2]

↳ magnitudes of accel.

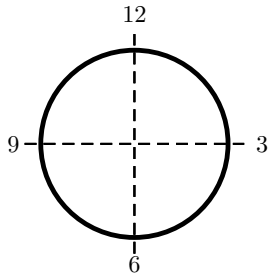
magnitude of $\vec{a} = \| \langle 6, -12 \rangle \|$

$= \sqrt{6^2 + (-12)^2} \approx 13.4$

Likeliest we are on Neptune.

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8. Assume time t runs from zero to 2π and that the unit circle has been marked with the hours as a clock face (see diagram at right). Match each of the parametric equations with the best description of the curve from the following list. For this question, no explanation or calculations are necessary.



A. Starts at 12 o'clock and moves clockwise one time around.

B. Starts at 12 o'clock and moves counterclockwise one time around.

C. Starts at 6 o'clock and moves counterclockwise one time around.

D. Starts at 3 o'clock and moves counterclockwise one time around.

E. Starts at 9 o'clock and moves counterclockwise one time around.

F. Starts at 9 o'clock and moves clockwise to 3 o'clock.

G. Starts at 9 o'clock and moves counterclockwise two times around.

H. Starts at 9 o'clock and moves clockwise two times around.

[/5]

(a) $\vec{r}(t) = \left\langle -\cos\left(\frac{t}{2}\right), \sin\left(\frac{t}{2}\right)\right\rangle$ matches description F

half angular velocity → only half circle.



(b) $\vec{r}(t) = \langle -\cos(2t), -\sin(2t)\rangle$ matches description G

two times angular velocity → twice around.



(c) $\vec{r}(t) = \langle \cos(t), \sin(t)\rangle$ matches description D



(d) $\vec{r}(t) = \langle \sin(t), \cos(t)\rangle$ matches description A



(e) $\vec{r}(t) = \langle -\cos(t), -\sin(t)\rangle$ matches description E



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Space for additional work. **Indicate clearly which question you are continuing if you use this space.**

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